EXPERIMENT 9

D15A-57

Aim: Design and train a Convolutional Neural Network (CNN) on image datasets

Theory:

1. Dataset Source
The dataset used is the Air Quality Dataset, available at: https://archive.ics.uci.edu/dataset/360/air+quality. It contains real-world air pollution measurements collected from sensors.

2. Dataset Description
The Air Quality dataset contains hourly averaged responses from an array of chemical sensors deployed in an Italian city. It consists of 9358 instances with 15 attributes, including measurements such as CO concentration, NOx levels, temperature, and relative humidity. The dataset also includes Date and Time columns indicating when each observation was recorded. Missing values are represented by -200. The data is sequential in nature, making it suitable for time series analysis and prediction tasks.

3. Mathematical Formulation of the Algorithm
A Long Short-Term Memory network is a type of Recurrent Neural Network designed to capture temporal dependencies. The LSTM unit consists of gates that control the flow of information:

Forget gate:

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f)$$

Cell state update:

$$C_t = f_t \cdot C_{t-1} + i_t \cdot \tilde{C}_t$$

Hidden state:

$$h_t = o_t \cdot \tanh(C_t)$$

These equations allow the model to retain long-term dependencies in sequential data.

4. Algorithm Limitations
LSTM networks require significant computational resources and training time. They are sensitive to hyperparameters such as sequence length, number of units, and learning rate. If the dataset is too noisy or contains many missing values, performance may degrade. LSTMs can also suffer from overfitting when trained on limited data. Additionally, they are less interpretable compared to traditional statistical models.

5. Methodology / Workflow

The dataset is first loaded and preprocessed by removing non-numeric columns such as Date and Time. Missing values represented by -200 are replaced with NaN and removed. The selected feature, such as CO concentration, is normalized to improve training efficiency. The data is then converted into sequential input-output pairs, where each value is used to predict the next time step. The sequences are formatted into cell arrays suitable for LSTM input. An LSTM network is defined with input, hidden, and output layers, followed by training using an optimizer such as Adam. After training, predictions are generated and compared with actual values using graphical visualization.

6. Performance Analysis

The performance of the model is evaluated by comparing predicted values with actual time series values. A plot is used to visualize how closely the predicted sequence follows the real data. A good model shows predictions that align closely with actual values. Metrics such as Mean Squared Error or Root Mean Squared Error can also be used to quantify prediction accuracy. The analysis helps determine how well the model captures temporal patterns in the data.

7. Hyperparameter Tuning

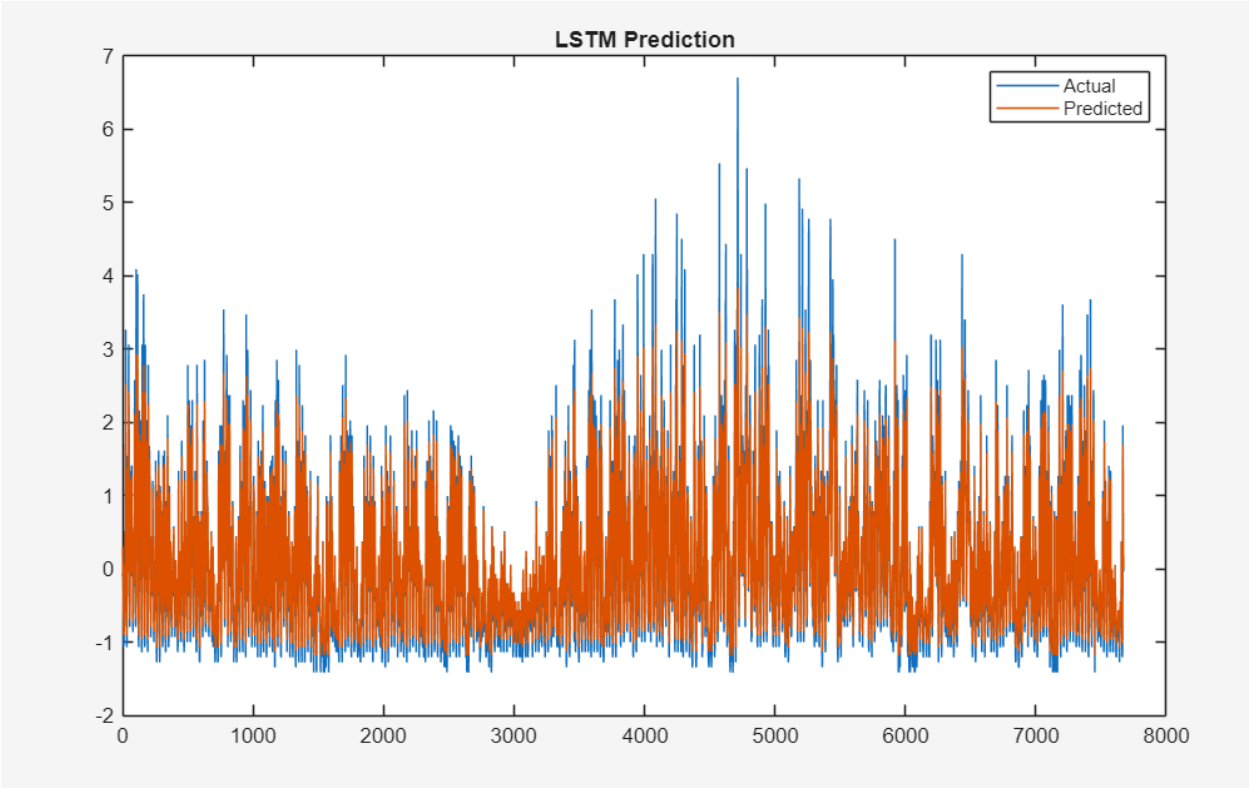
Hyperparameter tuning involves adjusting parameters such as the number of LSTM units, number of layers, learning rate, batch size, and number of epochs. Increasing the number of LSTM units can improve the model's ability to capture complex patterns, while adjusting the learning rate affects convergence speed. Batch size influences training stability, and the number of epochs determines how long the model learns from the data. By experimenting with different combinations, the model performance can be improved and better generalization can be achieved

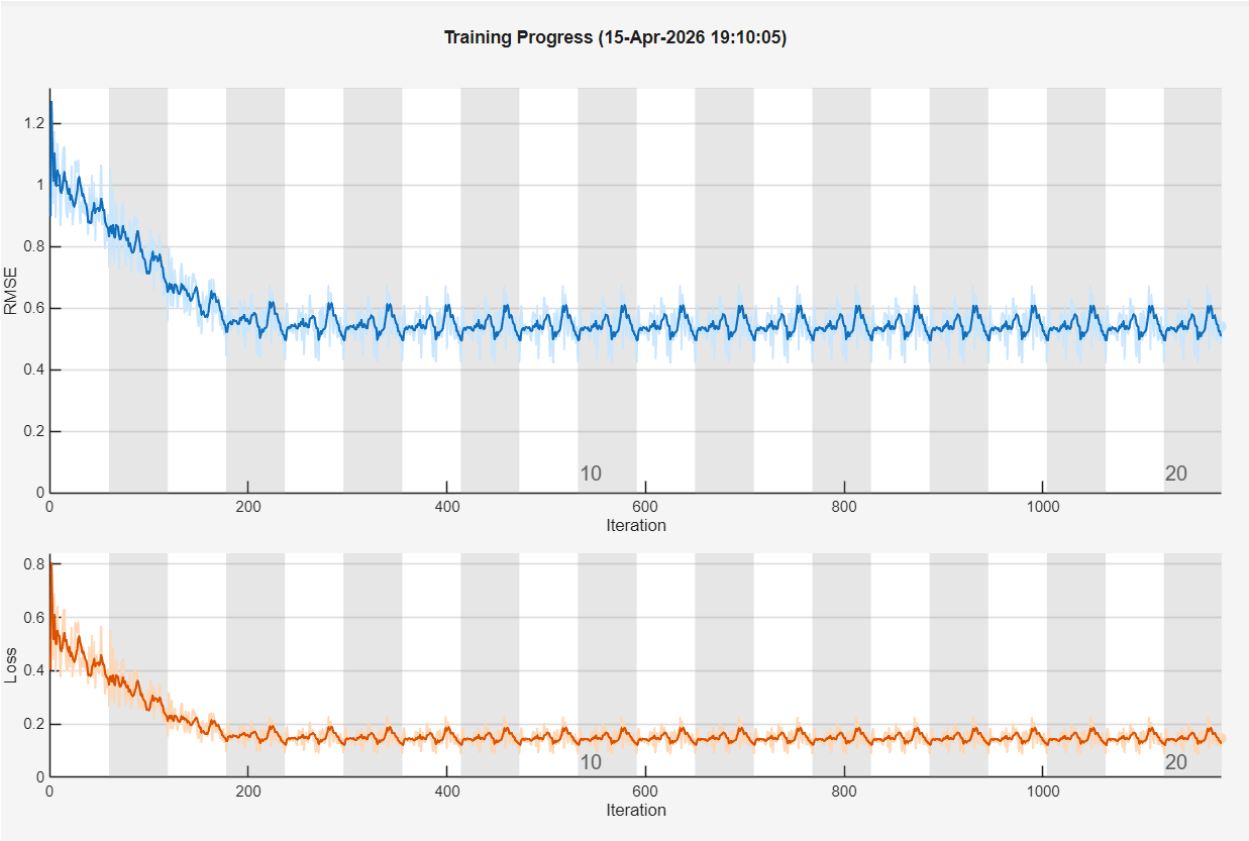
Code:

```
clc;
clear;
close all;

% Load dataset
data = readtable('AirQualityUCI.csv','VariableNamingRule','preserve');

% Remove Date and Time
data(:,1:2) = [];

% Convert table to cell
dataCell = table2cell(data);

% Convert all values to numeric
dataNum = nan(size(dataCell));

for i = 1:size(dataCell,1)
    for j = 1:size(dataCell,2)
        val = dataCell{i,j};

        if ischar(val) || isstring(val)
            val = strrep(val, ',', '.'); % replace comma
            num = str2double(val);
        else
            num = val;
        end

        dataNum(i,j) = num;
    end
end

% Replace -200 with NaN
dataNum(dataNum == -200) = NaN;
```

```
co = dataNum(:,1);
% Remove NaN
co = co(~isnan(co));
% Normalize
co = (co - mean(co)) / std(co);
% Prepare sequences
X = co(1:end-1);
Y = co(2:end);
XTrain = num2cell(X');
YTrain = num2cell(Y');

% LSTM layers
layers = [
    sequenceInputLayer(1)
    lstmLayer(50,'OutputMode','sequence')
    fullyConnectedLayer(1)
    regressionLayer
];
% Training options
options = trainingOptions('adam', ...
    'MaxEpochs',20, ...
    'MiniBatchSize',128, ...
    'Verbose',0, ...
    'Plots','training-progress');
% Train
net = trainNetwork(XTrain, YTrain, layers, options);
% Predict
YPred = predict(net, XTrain);
YPred = cell2mat(YPred);
% Plot
figure;
plot(co(2:end));
hold on;
plot(YPred);
legend('Actual','Predicted');
title('LSTM Prediction');
```

Output:


LSTM Prediction
Actual
Predicted
7
6
5
4
3
2
1
0
-1
-2
0
1000
2000
3000
4000
5000
6000
7000
8000


Training Progress (15-Apr-2026 19:10:05)
RMSE
1.2
1
0.8
0.6
0.4
0.2
0
10
20
0
200
400
600
800
1000
Iteration
Loss
0.8
0.6
0.4
0.2
0
10
20
0
200
400
600
800
1000
Iteration

Conclusion:

The implementation of a Long Short-Term Memory network on the Air Quality Dataset demonstrates the effectiveness of recurrent models in handling time series data. The model successfully captures temporal dependencies and predicts future values based on past observations. The results show that LSTM networks are well-suited for sequential data analysis, particularly in applications involving environmental monitoring. Although challenges such as missing data and parameter tuning exist, the approach provides reliable predictive performance and can be further enhanced with improved preprocessing and model optimization.